

Oleksii Odarchuk



Backend Developer (Go) · Creator of rombik and the Piton language

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Backend engineer (Go) with a product mindset and half a year of commercial experience. 1st at BEST Hackathon 2026, 3rd at Mate Hackathon, a personal invitation to DOU Day 2026. From hackathon MVPs to production Go services.

1st

place · BEST Hackathon 2026

10 langs

in my own rombik engine

1000×

speed-up (SHMiner)

150+

bot users (first year)

TECH STACK

Languages

GoPythonCTypeScript

Backend

GinFiberRESTWebSocketsSSEGoroutinesJWTOAuth 2.0

Data

PostgreSQLpgvectorRedisClickHouse

Frontend

SvelteWailsTailwind

AI

Anthropic ClaudeOpenAIMCPRAGfal.ai

DevOps

DockerKubernetesLinux (Arch)GitTinyGoTypst

EXPERIENCE

Backend Developer (Go)

October 2025 — present

SkyService

- Design and build **architecturally complex Go microservices from scratch** for a production platform: interface-driven (hexagonal) architecture, inter-service communication, work under real production load.
- Designed an AI orchestrator — a web analog of claude-code**: a control-plane / execution-plane split — Claude runs through a tool-use loop while **all agent code executes inside isolated Docker sandboxes**. Guarantee, **enforced by an automated test**: secrets (Anthropic key, git tokens) **never reach the sandbox**. MCP-over-SSE, credential-free git (GitHub App), an **egress-allowlist proxy** against exfiltration, a container-lifecycle supervisor.
- AI support-bot platform (RAG)**: semantic memory on PostgreSQL + pgvector (text / vector / hybrid search), a self-learning knowledge graph, an MCP server, schema-per-tenant isolation, token-based billing.

WINS & HACKATHONS

🏆 1st Place — Lead Backend & Sensor Fusion

April 18-19, 2026

BEST Hackathon 2026 — GPS-less Drone Trajectory

- Algorithm to **reconstruct a UAV's trajectory without GPS** from IMU/barometer logs. Sensor fusion in **Go + C**: Go for the pipeline & integration, C for the hot numerical kernels.
- 1st place among top teams** — judges singled out the architectural resilience: modular split telemetry / integration / simulation let us swap datasets without rewriting.

🏆 3rd Place — Backend Engineer

March 14, 2026

Mate Hackathon (50 participants, 10 teams)

- Designed and built a Go backend from scratch in **7 hours** for an AI-powered MarTech tool that automates competitor-ad analysis.
- Dual AI pipeline: GPT-4o for trigger analysis + fal.ai for creative generation; backend routed all AI calls under load. Beat teams with 5+ years of experience.

Security Researcher → Job Offer

2026

College Blockchain System “Student Hryvnia”

- Discovered and exploited a critical vulnerability: the **backend never checked transaction amounts** (only the frontend) → arbitrary negative-value transfers. Reported responsibly — received a job offer.
- Reverse-engineered the official JS miner via Webpack bundles — then a faster Go replacement (SHMiner — in Projects).

📅 **DOU Day 2026 — personal invitation.** Personally invited by the organisers to the conference with a free ticket — while still a student, ahead of professional work.

📁 PROJECTS

FLAGSHIP · COMMERCIAL micro-SaaS

[website](#) → [API](#) →

💎 rombik — Flowchart Generator

My first full-fledged commercial micro-SaaS · code in 10 languages → a DSTU 19.701-90 / ISO 5807 flowchart or structogram

Designed and run entirely by me. A **Go engine** — a compiler pipeline (parser → IR → layout → render) on **tree-sitter** for **10 languages** (Python, JavaScript, TypeScript, C, C++, C#, Java, Go, PHP, Pascal); flowcharts and Nassi-Shneiderman structograms, pixel-for-pixel to DSTU 19.701-90 / ISO 5807, byte-exact golden tests. Web editor (pan/zoom, lasso-splitting), 8 formats (SVG/PNG/PDF/Word·Visio·draw.io as native shapes/Typst/Excalidraw). **Billing runs on my own go-monobank-sdk** (HMAC webhooks, grants, receipts). Compiler craft carried over from my language **Piton**. Public **API, CLI, MCP server**, an AI skill and describe→flowchart via AI. Free watermarked PNG, a clean export needs an account.

Go · Gin

tree-sitter

API · CLI · MCP

go-monobank-sdk

PostgreSQL

DSTU

OPEN SOURCE SDK

go-monobank-sdk — Go SDK for every monobank API [github](#) →

Full coverage of **personal**, **provider** (with monoКЕП), **business** for legal entities, **acquiring** (subscriptions, monipay button, split-receivers, T2P) and **Installment Purchases**. ECDSA webhook verification with auto-rotating keys, drop-in `http.Handler`, typed money/currency/MCC, HMAC-SHA256 for installment callbacks, a fake monobank server for tests, and a separate OpenTelemetry sub-module.

Go

ECDSA

HMAC-SHA256

OpenTelemetry

iter.Seq2

SDK CONSUMER

monokasa — Telegram Ticket-Selling Bot [github](#) →

Sells tickets to a single show through a monobank jar and issues PDFs with QR codes scanned at the venue entrance. Live consumer of my own **go-monobank-sdk**: webhook auto-registers at startup, reservations get a 15-minute hold, and show-time reminders fire at-most-once even across restarts.

Go

SQLite

telebot

PDF + QR

go-monobank-sdk

1000× FASTER

SHMiner — Desktop Crypto Miner

[github →](#)

Cross-platform desktop miner built with **Wails (Go + Svelte/TypeScript)**. Rewrote SHA-256 hashing from JS to Go with a goroutine worker pool — achieved **1000× faster hashrate** than the official client on identical hardware. AES-encrypted wallet storage, real-time WebSocket dashboard, multi-wallet, zen mode.

[Go](#) [Svelte](#) [Wails](#) [WebSockets](#)
[Goroutines](#)

150+ USERS

AbitAssistant Bot

[github →](#)

Telegram bot for tracking university admission rankings in Ukraine. Reverse-engineered closed third-party APIs for live competition data. **150+ active users in the first year**, marketing campaign for 2026.

[Python](#) [Aiogram](#) [PostgreSQL](#)
[BeautifulSoup](#) [Pandas](#) [Docker](#)

TELEGRAM · ANALYTICS

LinkShortener Bot

[github →](#)

Telegram bot for shortening links with QR-code generation and geo-analytics. PostgreSQL for data, Redis for redirect caching, ClickHouse for real-time analytics (country, city, platform). Docker, GeolP2.

[Go](#) [PostgreSQL](#) [Redis](#) [ClickHouse](#)
[Docker](#) [GeolP2](#)

MY OWN LANGUAGE

Piton — My own programming language

[github →](#) [docs →](#)

Ukrainian transliterated language + interpreter, from scratch in **Go without yacc/bison**: recursive descent parser, tree-walking evaluator, Turing-complete. Flowcharts via a built-in **D2**-based engine. Interpreter compiled to **WebAssembly** via **TinyGo** (220 KB).

[Go](#) [TinyGo](#) [WebAssembly](#) [D2](#) [Nix](#)

🎓 EDUCATION & LANGUAGES

Network Technologies

IT College NULP, 2nd year
2024 — present

IT Class

Science Lyceum, Rivne
2022 — 2024

Ukrainian — native
English — B1

🎯 WHAT I'M LOOKING FOR

A backend role in **Go** focused on **designing systems**, not just CRUD: distributed services, developer tools, platform engineering, AI infrastructure. I value owning a product from idea to production — from the compiler to the billing — and working with engineers I can learn from.